COER UNIVERSITY

DBMS LAB ASSIGNMENT | LABSHEET 1:DATADEFINITION LANGUAGE (DDL)

Task 1: Database & Table Creation

Question 1.a

Question:CreateadatabasenamedCompany.

Answer:

```
-- Create a new database named Company
CREATE DATABASE Company;



-- Switch to the newly created database
USE Company;
```

Explanation: The CREATE DATABASE statement creates a database logical container named Company in MySQL. The USE statement selects Company as the active database for subsequent operations.

Question 1.b

Question: Create a table named Employees with the following structure: EmpID (INT, AUTO_INCREMENT, PRIMARY KEY), FirstName (VARCHAR(50), NOT NULL), LastName (VARCHAR(50)), Department (VARCHAR(50)), Salary (DECIMAL(10,2)), Email (VARCHAR(100), UNIQUE).

Answer:

```
-- Create Employees table with specified column constraints
CREATE TABLE Employees (

    EmpID INT AUTO_INCREMENT PRIMARY KEY,
    FirstName VARCHAR(50) NOT NULL,
    LastName VARCHAR(50),
    Department VARCHAR(50),
    Salary DECIMAL(10,2),
    Email VARCHAR(100) UNIQUE
);
```

Explanation: This statement defines the Employees schema. EmpID automatically increments for new rows and serves as the primary key. FirstName is mandatory (NOT NULL), and Email enforces uniqueness across records.

Task 2: DDL Queries on the Employees Table

Question 2.a

Question:Describe the structure of the table.

Answer:

```
-- View the structure and details of the Employees table
DESCRIBE Employees;
```

Explanation: The DESCRIBE command displays column names, data types, nullability, keys, default values, and auto-increment properties of the table.

Question 2.b

Question:Rename the table to Staff.

Answer:

```
-- Rename table Employees to Staff
RENAME TABLE Employees TO Staff;
```

Explanation: RENAME TABLE changes the table name from Employees to Staff in the database catalog without altering stored data or structure.

Question 2.c

Question:Adda new column PhoneNumber (VARCHAR(15)).

Answer:

```
-- Add PhoneNumber column to Staff table
ALTER TABLE Staff
ADD PhoneNumber VARCHAR(15);
```

Explanation: ALTER TABLE ... ADD appends a new column named PhoneNumber to the existing table definition.

Question 2.d

Question:Modify the data type of Salary to DECIMAL(12,2).

Answer:

```
-- Modify data type of Salary column
ALTER TABLE Staff
MODIFY Salary DECIMAL(12,2);
```

Explanation: ALTER TABLE ... MODIFY changes the column precision of Salary to support numbers up to 12 total digits with 2 decimal places.

Question 2.e

Question:Adda DEFAULT value of 'General' to the Department column.

Answer:

```
-- Set default value for Department column
ALTER TABLE Staff
ALTER Department SET DEFAULT 'General';
```

Explanation: Setting a DEFAULT constraint ensures that if a record is inserted without a department, MySQL automatically assigns 'General'.

Question 2.f

Question:Drop the PhoneNumber column.

Answer:

```
-- Remove PhoneNumber column from Staff table
ALTER TABLE Staff
DROP COLUMN PhoneNumber;
```

Explanation: ALTER TABLE ... DROP COLUMN permanently removes the PhoneNumber attribute from the table.

Question 2.g

Question:Adda CHECK constraint to ensure Salary is greater than 0.

Answer:

```
-- Add CHECK constraint on Salary column
ALTER TABLE Staff
ADD CONSTRAINT chk_Salary CHECK (Salary > 0);
```

Explanation: Adds a domain constraint named chk_Salary to validate that any inserted or modified salary value is strictly greater than 0.

Question 2.h

Question:Rename the column FirstName to GivenName.

Answer:

```
-- Rename column FirstName to GivenName
ALTER TABLE Staff
RENAME COLUMN FirstName TO GivenName;
```

Explanation: RENAME COLUMN changes the column identifier from FirstName to GivenName while maintaining its data type and constraints.

Question 2.i

Question:Drop the chk_Salary constraint.

Answer:

```
-- Drop CHECK constraint from Staff table
ALTER TABLE Staff
DROP CHECK chk_Salary;
```

Explanation: Removes the chk_Salary check constraint from the table definition.

Question 2.j

Question:Create an index on the Email column.

Answer:

```
-- Create non-unique index on Email column
CREATE INDEX idx_email ON Staff(Email);
```

Explanation: Creates a B-tree index named idx_email on the Email column to speed up search and query operations.

Question 2.k

Question:Drop the index on the Email column.

Answer:

```
-- Drop index idx_email from Staff table
DROP INDEX idx_email ON Staff;
```

Explanation: DROP INDEX deletes the specified index structure from the table.

Question 2.l

Question:Truncate the table to remove all rows without affecting the structure.

Answer:

```
-- Truncate Staff table to delete all rows
TRUNCATE TABLE Staff;
```

Explanation: TRUNCATE TABLE removes all records from the table quickly and resets auto-increment counters, while keeping the table structure intact.

Question 2.m

Question:Dropthe Staff table completely.

Answer:

```
-- Permanently drop Staff table
DROP TABLE Staff;
```

Explanation: DROP TABLE deletes the table definition, data, constraints, and indexes permanently from the database.

Task 3: Conceptual Questions

Question 3.1

Question:What is the difference between TRUNCATE and DELETE?

Answer:

1. Command Category: TRUNCATE is a DDL command, whereas DELETE is a DML command.
2. Speed & Efficiency: TRUNCATE is faster as it deallocates memory pages directly; DELETE removes rows one by one.
3. WHERE Clause: DELETE supports conditions using WHERE; TRUNCATE removes all rows unconditionally.
4. Auto-increment Reset: TRUNCATE resets AUTO_INCREMENT counters back to 1; DELETE retains the current counter value.

Explanation: Understanding the distinction helps in choosing the right command based on performance needs and transaction safety.

Question 3.2

Question:Explain the use of the CHECK constraint.

Answer:

The CHECK constraint enforces domain integrity by ensuring that all values entered in a column satisfy a logical boolean condition (e.g., Salary > 0 or Age >= 18). If an insert or update violates the condition, MySQL throws an error and rejects the operation.

Explanation: CHECK constraints prevent invalid data entry at the database level.

Question 3.3

Question:Whyis it important to define a PRIMARY KEY for a table?

Answer:

1. Uniqueness: Guarantees that every row in the table can be uniquely identified.
2. Entity Integrity: Ensures key columns cannot contain NULL values.
3. Performance: MySQL creates a clustered index on the primary key for fast row lookup.
4. Relationships: Acts as a reference key for foreign keys in other tables.

Explanation: Primary keys maintain fundamental database integrity and optimize search queries.

Question 3.4

Question:Howdoes ALTER TABLE help in modifying an existing table?

Answer:

ALTER TABLE enables structural changes to an existing table without deleting stored data. It supports adding or dropping columns, changing data types, modifying constraints, setting default values, and managing indexes.

Explanation: It allows database schemas to evolve over time without losing existing records.

Question 3.5

Question:What is the purpose of creating an index in a table?

Answer:

Anindexbuilds asearchpointer structure (B-tree) onspecifiedcolumns to significantly accelerate record retrieval during SELECT queries (WHERE, JOIN, ORDER BY), eliminating the need for full table scans.

Explanation: Indexes trade small storage space and insert overhead for much faster reading performance.

Task 4: Exercises

Question 4.1

Question: Create a table named Projects (ProjectID INT PRIMARY KEY AUTO_INCREMENT, ProjectName VARCHAR(100) NOT NULL, StartDate DATE, EndDate DATE, Budget DECIMAL(15,2) CHECK (Budget > 0)). Perform at least 5 DDL operations on Projects, and write a query to retrieve column names.

Answer:

```
CREATE TABLE Projects (
    ProjectID INT PRIMARY KEY AUTO_INCREMENT,
    ProjectName VARCHAR(100) NOT NULL,
    StartDate DATE,
    EndDate DATE,
    Budget DECIMAL(15,2) CHECK (Budget > 0)
);

ALTER TABLE Projects ADD Location VARCHAR(100);

ALTER TABLE Projects MODIFY ProjectName VARCHAR(150);

ALTER TABLE Projects RENAME COLUMN Location TO ProjectCity;

CREATE INDEX idx_proj_name ON Projects(ProjectName);

DROP INDEX idx_proj_name ON Projects;

-- Query to retrieve column names
SELECT COLUMN_NAME
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_NAME = 'Projects' AND TABLE_SCHEMA = 'Company';
```

Explanation: Creates the Projects table with constraints, executes 5 DDL operations (ADD, MODIFY, RENAME COLUMN, CREATE INDEX, DROP INDEX), and queries system metadata to display table columns.